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MILITARY SPECIFICATION

COMPRESSOR UNIT, AIRCRAFT, ELECTRIC MOTOR DRIVEN,
GENERAL SPECIFICATION FOR

This specification was approved by the Departments of the Army, the Navy, and the Air Force for use of procurement services of the respective Departments.

1. SCOPE

1.1 This specification covers electric motor-driven air compressors used to supply high pressure air for pneumatic actuating systems.

2. APPLICABLE SPECIFICATIONS, OTHER PUBLICATIONS, AND DRAWINGS

2.1 The following publications, of the issue in effect on date of invitation for bids, shall form a part of this specification to the extent specified herein:

2.1.1 Specifications.-

Federal

QQ-P-416	Plating, Cadmium (Electrodeposited)

Military

MIL-D-5028	Drawings and Data Lists; Preparation of (For Engines, Accessories and Other Auxiliary Equipment)
MIL-H-5440	Hydraulic Systems; Design, Installation and Tests of Aircraft (General Specification for)
MIL-I-6868	Inspection Process, Magnetic Particle
MIL-L-6880	Lubrication of Aircraft, General Specification for
MIL-M-3171	Magnesium Alloy; Processes for Corrosion Protection of
MIL-M-7911	Marking, Identification of Aeronautical Equipment, Assemblies, and Parts
MIL-P-5514	Packings; Installation and Gland Design of Aircraft Hydraulic and Pneumatic (General Specification for)
MIL-P-5633	Packaging and Packing of Aircraft Material in Steel Shipping Containers
MIL-P-6871	Plating, Chromium
MIL-S-7742	Screw Threads, Standard, Aeronautical

Air Force-Navy Aeronautical

AN-QQ-A-696	Anodic-Films; Corrosion-Protective (for) Aluminum Alloys

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AN-M-40	Motors; Aircraft Direct Current, 24 Volt System
AN-P-32	Plating; Zinc

U. S. Air Force

32590	Motors; Aircraft, Alternating Current, 400 Cycle, Induction, General Specification for

2.1.2 Other Publications.-

Military Standard (Book)

MIL-STD-10	Surface, Roughness, Waviness, and Lay

Air Force-Navy Aeronautical Bulletin

No. 143	Specifications and Standards; Use of

2.1.3 Drawings.-

Air Force-Navy Aeronautical Standard Drawings

AN814	Plug and Bleeder - Screw Thread
AN995	Wire - Lock
AN6290	Gasket - Straight Thread Tube Fitting, Boss
AND10050	Bosses - Standard Dimensions for Gasket Seal Straight Thd
AND10074	Boss Spacing - Hydraulic
AND10260	Flange - Type X Accessory Mounting
AND10261	Flange - Type XI Accessory Mounting
AND10262	Flange - Type XII Accessory Mounting
AND10398	Metals - Definition of Dissimilar
AND20000	Drive - Type X Engine Accessory
AND20001	Drive - Type XI Engine Accessory
AND20002	Drive - Type XII Engine Accessory

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Components.- The electric motor-driven air compressor shall consist of a compressor and an electric motor connected together with a drive and flange assembly, as specified herein, an attached pressure switch, and a fan for cooling if required.

3.2 Materials.- Materials shall conform to applicable specifications and shall be as specified herein. Materials which are not covered by applicable specifications, or which are not specifically described herein, shall be of the best quality, of the lightest practicable weight, and suitable for the purpose intended.

3.2.1 Metals.- All metals used in the construction of electric motor-driven air compressors, except metal which is in constant contact with oil, shall be of a corrosion-resistant type or shall be suitably protected to resist corrosion.

3.2.2 Dissimilar Metals.- Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in Drawing AND10398.

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3.2.3 Plastic Parts.- The use of plastic parts shall be subject to the approval of the Services for the specific application involved.

3.2.4 Selection of Materials.- Specifications and standards for all materials, parts, and Government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with ANA Bulletin No. 143, except as provided in the following paragraph.

3.2.4.1 AN or JAN Standard Parts.- AN or JAN Standard parts shall be used wherever they are suitable for the purpose, and shall be identified on the drawing by their part numbers. Commercial utility parts such as screws, bolts, nuts, cotter pins, etc., may be used, provided they possess suitable properties and are replaceable by the AN or JAN Standard parts without alteration, and provided the corresponding AN or JAN part numbers are referenced in the parts list and, if practicable, on the contractor's drawings. In the event there is no suitable corresponding AN or JAN Standard part in effect on date of invitation for bids, commercial parts may be used provided they conform to all requirements of this specification.

3.3 Design and Construction.-

3.3.1 General.- The configuration, dimensions, and other details of design shall conform to the requirements of the applicable AN or MS Standard drawings.

3.3.2 Weight.- Weight shall be kept to a minimum consistent with the requirements of this specification. The weight of the assembled component shall be specified on the manufacturer's assembly drawings.

3.3.3 Temperature Range.- The component shall be designed to operate satisfactorily throughout a temperature range of -54° to 71°C (-65° to 160°F).

3.3.4 Direct-Current Motors.- All direct-current, 24-volt system, electric motors shall be in accordance with Specification AN-M-40. (Where duplication of tests occurs between tests in this specification and those in Specification AN-M-40, the tests in this specification shall suffice.)

3.3.5 Alternating-Current Electric Motors.- All 400-cycle, alternating-current, electric motors shall be in accordance with Specification 32590. (Where duplication occurs between tests in this specification and those in Specification 32590, the tests in this specification shall suffice.)

3.3.6 Drive and Flange.- The electric motor shall have a drive conforming dimensionally to Drawing AND20000, AND20001, or AND20002. The air compressor shall have a flange which mates with the corresponding drive of the electric motor, and conforming to Drawing AND10260, AND10261, or AND10262. The above shall be complied with unless it can be shown to the satisfaction of the procuring agency that appreciable advantages are to be gained by other drive configurations.

3.3.6.1 Gasket.- A mounting flange gasket, as specified on the applicable AND drive drawing, shall be incorporated between the pump and motor.

3.3.7 Balance.- The moving parts of the electric motor-driven air compressor shall be balanced and shall not vibrate excessively under any conditions of operation.

3.3.8 Cooling Air.- The compressor shall provide its own cooling fan if required.

3.3.9 Mounting Position.- Mounting position shall be as specified by the procuring agency. The compressor shall withstand an inverted flight attitude for 1 minute.

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3.3.10 Lubrication.- The compressor shall be lubricated from its own oil supply. Lubrication shall be in accordance with Specification MIL-L-6880. The oil and lubrication system shall enable satisfactory performance between temperatures of -65° to 165°F. Replenishment of the oil shall not be necessary for at least 50 hours of operation. A convenient means shall be provided for refilling of the reservoir and determining the level of oil at all times.

3.3.11 Pressure Switch.- The pressure switch shall sense line pressure to the receiver and control the operation of the motor. The switch shall be adjustable within ranges specified by the procuring agency, in such a manner that the motor starts when the line pressure drops to some limiting value and stops when the line pressure rises to another limiting value. The sensitivity of adjustment shall be such that all starts shall occur within ±20 psi of the specified value and all stops within ±20 psi of the specified value.

3.3.12 Plugs.- All plugs, except permanently installed plugs which will not have to be removed during the life of the component, shall conform to Drawing AN814, and shall be sealed with gaskets conforming to Drawing AN6290. Permanently installed plugs may be pipe-threaded or of any other form suitable for the purpose.

3.3.13 Threads.- Except for permanently installed pipe-threaded plugs, and unless otherwise specified on the applicable drawing, only straight threads conforming to Specification MIL-S-7742, National Fine Thread Series, class 3 (NF3), shall be used.

3.3.14 Bosses.- All bosses for connecting fittings shall conform to Drawing AND10050. Boss spacing shall comply with Drawing AND10074. Ports shall be permanently marked to indicate proper connections to be made.

3.3.15 Seals.- All packings and gaskets, and installations thereof, shall conform to the requirements of Specification MIL-P-5514.

3.3.16 Safetying.- All threaded parts shall be positively locked or safetied by safety-wiring, self-locking nuts, or other approved methods. Safety wire shall have a minimum diameter of 0.032 inch and shall conform to Drawing AN995.

3.3.17 Snap Rings.- Snap rings, if used, shall be used in accordance with the requirements of Specification MIL-H-5440.

3.4 Interchangeability.- All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part numbers shall be governed by the drawing number requirements of Specification MIL-D-5028. Subassemblies, composed of selected mating components, must be interchangeable as assembled units, and shall be so indicated on the manufacturer's drawings. The individual components of such assembled units need not be interchangeable.

3.5 Finish.-

3.5.1 Surface Roughness.- Surface roughness finishes shall be established and shall be specified in the manufacturer's drawings as outlined in MIL-STD-10. The determination of surface finish shall be made by a profilometer, comparator, brush analyzer, or other suitable comparison equipment with an accuracy of ±5 microinches at the level being measured.

3.5.2 Protective Treatment.- When materials are used in the construction of the pumps that are subject to corrosion in salt air or other atmospheric conditions likely to occur during service usage, they shall be protected against such corrosion in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of atmospheric conditions shall be avoided.

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3.5.3 Steel Parts.- Unless otherwise authorized, all steel or copper-alloy parts shall be cadmium-plated in accordance with Specification QQ-P-416, type I, class B; zinc-plated in accordance with Specification AN-P-32; or chrome-plated in accordance with Specification MIL-P-6871.

3.5.4 Aluminum-Alloy Parts.- Unless otherwise authorized, all aluminum-alloy parts shall be covered with an anodic-film conforming to Specification AN-QQ-A-696.

3.5.5 Magnesium-Alloy Parts.- Wherever practicable, magnesium-alloy parts shall be treated in accordance with Specification MIL-M-3171, to provide protection against corrosion.

3.5.6 No finishes or paints, other than those specified above, or color markings specified herein or otherwise authorized by the Services, shall be applied to the components either externally or internally prior to installation in the airplane.

3.6 Radio Noise Filter.- A radio noise filter, meeting the requirements of Specification AN-M-40, shall be incorporated in all direct-current units.

3.7 Magnetic Inspection.- All magnetizable highly stressed parts shall be subjected to magnetic inspection in accordance with Specification MIL-I-6868. Where necessary, such inspection shall be designated on the manufacturer's drawing. Cracks or other injurious defects disclosed by maghetic inspection shall be cause for rejection.

3.8 Performance.- The air compressor shall satisfy the performance requirements specified in Section 4, when subjected to the tests headed as follows:

(a) Examination of Product
(b) Break-In Run
(c) Rated Voltage Run
(d) Capacity (paragraph 4.4.2.4)
(e) Capacity (paragraph 4.4.3.2)
(f) Endurance
(g) Operation
(h) High Temperature
(i) Low Temperature Starting

3.9 Special Tools.- The air compressor shall be so designed that special tools will not be required for assembly or disassembly.

3.10 Identification of Product.-

3.10.1 Nameplate.- A nameplate, permanently and legibly filled in with the information listed below shall be securely attached to the compressor. The information marked in the spaces provided shall be in accordance with Specification MIL-M-7911.

COMPRESSOR UNIT, AIRCRAFT, ELECTRIC MOTOR DRIVEN
Rated Pressure
Rated Voltage (AC or DC)
Rated Amperage
Cycles AC (if applicable)
Specification MIL-C-6591A
Stock No.
Manufacturer's Part No.
Manufacturer's Serial No.
Contract or Order No.
Manufacturer's Name or Trade-Mark
U S Property

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3.11 Workmanship.- All details of workmanship shall be in accordance with high-grade aircraft electric motor and air compressor manufacturing practice.

3.11.1 Cleaning.- All parts shall be clean and free from dirt, sand, metal chips, and other foreign matter, while being assembled and after assembly.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Classification of Tests.- The inspection and testing of air compressors shall be classified as follows:

(a) Preproduction tests: Preproduction tests are those tests accomplished on samples which are representative of the production of the item after the award of contract, to determine that the production compressor meets the requirements of this specification.

(b) Inspection tests: Inspection tests are those tests accomplished on compressors manufactured and submitted for acceptance under contract.

4.2 Test Conditions.-

4.2.1 Cleaning.- All oil and grease or other corrosion-resisting compounds shall be removed from the interior and exterior parts of the compressor and external parts of the electric motor.

4.2.2 Oil.- Unless otherwise specified, all tests shall be conducted using oil conforming to a Government specification, to be specified by the procuring agency.

4.2.3 Measurements and Apparatus.-

4.2.3.1 Inlet Pressure.- The inlet pressure shall be measured with a calibrated manometer or equally accurate pressure gage located as close to the inlet port as practicable.

4.2.3.2 Outlet Pressure.- The outlet pressure shall be measured with a calibrated pressure gage located as close to the discharge port as practicable.

4.2.3.3 Motor Voltage.- The input voltage shall be measured at the terminal of the motor by means of a precision laboratory-type instrument accurate to 1 percent.

4.2.3.4 Motor Current.- The motor current shall be measured by a precision laboratory-type instrument accurate to 1 percent.

4.2.3.5 Electric Power.- The input power to an alternating-current motor shall be measured by a precision laboratory-type instrument accurate to 1 percent.

4.2.4 Filtration.- A filter shall be installed at the inlet for all tests.

4.2.5 Ambient Conditions.- Unless otherwise specified, the ambient pressure shall be 25 to 30 inches of mercury absolute and the ambient temperature shall be 70° to 100°F. No external forced circulation of air shall be allowed other than that which is built into the compressor unit.

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4.3 Preproduction Tests.-

4.3.1 Sampling Instructions.- One compressor of each manufacturer's part number shall be subjected to preproduction testing by the manufacturer. An opportunity shall be provided for a Government Inspector to witness such tests. The compressor shall be submitted to the cognizant Service with a complete test report in duplicate. This report shall contain original data sheets or photostatic copies thereof.

4.3.1.1 Detailed Information.- Information on performance requirements shall be specified as follows:

(a) Application (Brief description)

(b) Design

(1) Dimensions: The over-all dimensions, the size and location of compressor ports, and the type and location of electrical connectors shall be as shown on attached figure_____.

(2) Weight: The maximum dry weight of the complete compressor motor assembly shall not exceed _____ pounds.

(3) Rated Pressure: The rated delivery pressure of the compressor shall be _____ psi gage.

(4) Rated Capacity: The compressor capacity shall be at least _____ pounds per minute at rated input motor voltage, rated delivery pressure, 100 ±5°F air inlet temperature, and 20 inches of mercury absolute inlet pressure.

(5) Inlet Pressure: The minimum inlet pressure shall be _____ inches Mercury absolute, at which pressure the capacity shall be not less than _____ pounds per minute.

(6) Input Voltage: The normal input voltage when measured at the motor input terminals shall be _____.

(7) Duty Cycle: The maximum duty cycle of the unit shall be as follows:

_____

_____

(8) Explosion Proof: The unit _____ (shall or shall not) be explosion proof, as defined in Specification AN-M-40.

(9) Oil Proof: The unit _____ (shall or shall not) be oil proof, as defined in Specification AN-M-40.

(10) Thermal Protection: Thermal protection, as defined in Specification AN-M-40, _____ (shall or shall not) be required.

(11) Current: The maximum continuous running current at rated voltage, rated delivery pressure, and any

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condition of inlet pressure or temperature specified herein shall not exceed _____ amperes. The maximum starting current at the above determined conditions shall not exceed _____ amperes.

(12) Performance Curves: A graph showing compressor capacity (lb/min) versus inlet pressure (in.Hg abs) in increments of 5,000 feet from sea level to the maximum altitude specified by the procuring agency, shall be submitted. Each graph shall show curves corresponding to 90-, 100-, and 110-percent rated voltage.

(c) Additional Detail Requirements: (If available)

All requests for preproduction approval shall be accompanied by one complete set of detail and assembly, Vandyke, or other reproducible drawings, and one set of reproductions of these drawings, and two copies of the Preproduction test report with copies of the original test data sheets.

4.3.1.2 In the case of series of compressors which are intended to serve the same general function, approval of one compressor of the series may, at the discretion of the Services, be applied to any other compressor of the series if all of the internal working parts are identical in every detail with the corresponding internal working parts of the approved compressor and provided it meets such operational requirements as may be designated by the Services. For example, approval of this type would apply to all compressors which differ from previously approved ones only insofar as port size, port location, external body dimensions, and external body configuration are concerned.

4.3.1.3 Approval applies only to the design, materials, construction, and nominal dimensions of the compressor tested. Such features shall be identified by the manufacturer's part number submitted in his test report. Any change in any of these features may require new Preproduction tests at the option of the qualifying agency. Such changes shall be identified by the assignment of new numbers to the manufacturer's drawings of the component or its component parts, or both, and shall receive Government approval prior to incorporation in production. At the option of the Government, minor changes in these features may be approved without an attendant change in the part number. The contractor shall submit to the procuring agency a list of all parts which make up the compressor, giving drawing numbers with dated revisions at the time of submittal for tests. This list shall be revised and resubmitted when any changes are incorporated.

4.3.1.4 Drawings.- The assembly drawings shall show a cutaway section showing all details in their normal assembly position and shall carry part numbers of all details and subassemblies. The following data shall be furnished on or together with all assembly drawings:

(a) Mounting Dimensions
(b) Port Dimensions
(c) Over-All Dimensions
(d) Rated Capacity
(e) Rated Pressure
(f) Rated Voltage
(g) Rated Current
(h) Weight

4.3.2 Tests.- Preproduction tests of air compressors shall consist of the tests specified under Inspection tests, and in addition, the ones described herein.

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4.3.2.1 Operation.- The capacity of the compressor electric motor input power and current shall be determined at the conditions specified below. The capacity shall be as specified in the procuring agency's detail requirements, paragraph 4.3.1.1, (b), (4). The compressor shall operate satisfactorily under all the conditions specified. The compressor inlet temperature shall be maintained at 100° ±5°F, unless otherwise specified.

4.3.2.1.1 Sea Level Operation.- Runs shall be made at 90-, 100-, and 110-percent rated voltage at 0- (or lowest possible pressure), 25-, 50-, 75-, and 100-percent rated delivery pressure, compressor inlet pressure 20 inches of mercury absolute, and compressor inlet temperature of 100° ±5°F. The compressor shall operate satisfactorily during these runs.

4.3.2.1.2 Altitude Operation.- The compressor capacity shall be determined at rated discharge pressure 90-, 100-, and 110-percent rated voltage and at various pressure altitudes. The ambient pressure in 5,000-foot increments, and the ambient and inlet temperatures shall be simulated as in table I from sea level to the maximum altitude specified by the procuring agency. At the option of the procuring agency, the inlet pressure may be either that selected to simulate pressurized inlet or the same as the ambient pressure in table I. During this test, the capacity shall fall within the limits of the curves submitted under paragraph 4.3.1.1, (b), (12). If the compressor inlet is to be connected to a source of pressurization, the above shall be modified to determine if the capacity falls within the range of inlet pressures as set forth by the procuring agency.

TABLE I

Altitude	Pressure in. Hg	Temperature °F
Sea Level	29.92	
5,000	24.89	
10,000	20.58	75 ±15
15,000	16.88	(Applies to all
20,000	13.75	altitudes and
25,000	11.10	pressures)
30,000	8.90	
35,000	7.00	
40,000	5.50	
45,000	4.00	

4.3.2.2 High Temperature.- The compressor assembly in a static condition shall be subjected to an ambient temperature of 160° to 165°F for a period of 30 hours. The compressor shall then be operated at rated voltage, rated delivery pressure, and 20 inches of mercury absolute inlet pressure for 1/2 hour. The compressor shall operate satisfactorily during this run.

4.3.2.3 Low Temperature Starting.- The electric motor-driven compressor shall be cooled by some suitable means to a temperature of -60° to -70°F, and shall be allowed to soak at this temperature for at least 72 hours. The compressor shall be started and allowed to pump a receiver at rated pressure. The input power and current shall be determined at 90-percent rated voltage, rated pressure, -65° ±5°F air inlet temperature, and 20 inches of mercury absolute air inlet pressure. This test shall be repeated 10 times, with a waiting period of at least 3 hours between each start. The compressor shall operate satisfactorily during this run.

4.3.2.4 Pressure Control.- This test may be conducted in conjunction with other Preproduction tests. The pressure switch shall be actuated for at least 1,000 on-off cycles and shall operate as specified in paragraph 3.3.11. This test shall be conducted at high and low temperatures, as specified in paragraphs 4.3.2.2 and 4.3.2.3.

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4.4 Inspection Tests.- The contractor shall furnish all samples and shall be responsible for accomplishing the required tests. When inspection is conducted at the contractor's plant, all inspection and testing shall be under the supervision of the Government Inspector. Contractors not having laboratory testing facilities satisfactory to the Government shall engage the services of a commercial testing laboratory acceptable to the procuring agency. The contractor shall furnish test reports showing quantitative results for all tests required by this specification and signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during course of manufacture shall in no case be construed as a guaranty of the acceptance of the finished product.

4.4.1 Tests.- Inspection of compressors shall consist of the following tests, subject to the test conditions specified herein.

4.4.2 Individual Tests.-

4.4.2.1 Examination of Product.- Each compressor shall be examined to determine conformance with the applicable drawings and all requirements of this specification for which there are no specific tests.

4.4.2.2 Break-In Run.- The break-in run shall be made with any desired pressure in the compressor inlet and outlet lines and shall be run at 90- to 110-percent rated voltage. The time of break-in run shall not exceed 2 hours. The compressor shall operate satisfactorily during this test and those following.

4.4.2.3 Rated Voltage Run.- Each compressor shall operate satisfactorily at rated voltage ±1 percent and rated pressure ±3 percent as follows. This test may be conducted as part of the Break-In Run in paragraph 4.4.2.2.

(a) One-half hour or the equivalent number of duty cycles at 18 to 20 inches mercury absolute inlet pressure.

4.4.2.4 Capacity.- At rated voltage ±1 percent, the current and capacity shall be measured and shall fall within the requirements specified in paragraph 4.3.1.1, (b), (4).

4.4.3 Sampling Tests.- One air compressor from the first sublot of 50, an additional one from the next sublot of 500, and also one from every sublot of 500 or fraction thereof on the order after the first 500 compressors, shall be selected at random and subjected to the following Sampling tests. These tests shall be in addition to the Individual tests specified herein.

4.4.3.1 Low Temperature Starting.- This test shall be conducted in accordance with paragraph 4.3.2.3.

4.4.3.2 Capacity.- The capacity of the compressor shall be measured at 100° ±5°F inlet temperature after equilibrium is reached under each of the conditions shown in table II. The capacity shall be at least that specified in paragraph 4.3.1.1, (b), (4).

TABLE II

Motor Voltage- Percent of Rated	Compressor Inlet Pressure Inches of Mercury Absolute	Compressor Delivery Pressure in Pounds per Square Inch +200, -0
90	20 ±5	100-percent rated
100	20 ±5	100-percent rated
110	20 ±5	100-percent rated

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4.4.3.3 Endurance.- The compressor shall satisfactorily complete 500 hours of operation as shown in table III with no failure of parts of excessive wear.

TABLE III

No. of Hours Operation	No. of Cycles 1/ Operation	Motor Terminal Voltage, Percent of Rated Voltage ±2 Percent	Inlet Temp °F	Inlet Pressure in. Hg abs
25	25	90	75 ±15	25 - 30
50	50	100	75 ±15	25 - 30
25	25	110	75 ±15	25 - 30
125	125	90	75 ±15	19.5 - 20.5
150	150	100	75 ±15	19.5 - 20.5
125	125	110	75 ±15	19.5 - 20.5

1/ One cycle shall consist of starting the compressor, allowing it to pressurize a receiver of no more than 100 cu in. from 90-percent rated pressure to rated pressure, continuing the operation at this pressure for a total running time of 30 minutes, shutting off for 30 minutes, and bleeding the receiver.

4.4.3.3.1 Starting.- During the Endurance test, continuous duty motors shall be stopped and started against 100-percent rated delivery pressure at least twice every 24 hours. The peak starting current shall be recorded at the start and finish of the Endurance test, and shall not exceed the value specified in paragraph 4.3.1.1, (b), (11).

4.4.3.3.2 Measurements shall be taken before and after the Endurance test, and a wear analysis shall be made showing the wear on each rubbing part. If, during the Endurance test, a major part fails, the test shall be terminated. The replacement compressor unit shall have a redesigned part or one of different material corresponding to the failed part except that, if the failure was caused by faulty material or workmanship, the procuring agency may authorize the installation of a part of the original design and material with the defect overcome. The Endurance test shall be considered completed when every major part has concurrently completed the Endurance test. Minor parts may be replaced and penalty runs conducted at the option of the procuring agency.

4.4.3.3.3 Recalibration.- Following the Endurance test, the capacity shall be determined in accordance with paragraph 4.4.3.2 entitled Capacity. The compressor capacity shall not have decreased more than 15 percent from the capacity prior to the Endurance test at the condition defined.

4.5 Rejection and Retest.- When tests are specified on an air compressor that is selected as representative of a certain lot, and this compressor fails to meet the requirements, at least three additional compressors of the sublot presented shall be tested immediately. In the event any of the additional samples fail to meet the requirements, the entire sublot shall be rejected. After the failure of any sample to pass any of the tests, subsequent production shall be sampled as though it were initial production. Individual performance tests shall not be interrupted, unless the defect is of such a nature that it will seriously affect the performance or safe use of the compressors. Rejected compressors shall not be resubmitted for inspection without full particulars concerning previous rejection and measures taken to overcome the defects.

4.6 All parts, specimens, or assemblies destroyed in making tests required by this specification or drawings, to determine compliance with the specification or drawings, shall be in addition to the quantity specified in the contract or purchase order and shall be furnished without increasing the cost of the contract or order.

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5. PREPARATION FOR DELIVERY

5.1 Packaging and Packing.- Each air compressor shall be packaged and packed for shipment in accordance with Specification MIL-P-5633.

5.2 Marking and Labeling.- Marking shall be as specified in Specification MIL-P-5633. In addition, each shipping container shall be marked with the following information:

COMPRESSOR UNIT, AIRCRAFT, ELECTRIC MOTOR DRIVEN
Rated Pressure
Rated Voltage (AC or DC)
Rated Amperage
Specification MIL-C-6591A
Stock No. (USAF procurement only)
Manufacturer's Part No.
Manufacturer's Serial No.
Contract or Order No.
Date of Manufacture
Quantity
Manufacturer's Name or Trade-Mark
Name of Contractor (if different from manufacturer)

IF IN STORAGE AFTER (DATE)*, THE COMPRESSOR SHALL BE TESTED AND INSPECTED BEFORE ISSUE.

*NOTE: This date shall be 18 months from the date of inspection, and shall be inserted by the Inspector at the time of acceptance.

6. NOTES

6.1 Intended Use.- The air compressors covered by this specification are intended for use in aircraft for pneumatic operation of such mechanisms as landing gears, brakes, flaps, gun turrets, bomb bay doors, and other intermittently actuated devices.

6.2 Drawing Requirements.- The attention of contractors is invited to the fact that Government contracts for equipment, whether procured directly by the Government or through another contractor, require that all drawings submitted, must be in accordance with the requirements of Specification MIL-D-5028.

6.3 Ordering Data.- Invitations for bids, contracts, and purchase orders shall state the conditions for the following.

6.3.1 Preproduction Tests.- It is expected that the contract or purchase order will specify that one electric motor-driven air compressor will be required as a preproduction sample and that this preproduction sample will be subjected to the Preproduction tests to determine compliance with the requirements of this specification. The invitation for bids and the contract should specify the point of inspection for these tests.

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